[TO BE COMPLETED: Paper Title — descriptive, technique + domain + dataset]

Preliminary version — Stage 1: state of the art and proposed design

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***Abstract*—Preliminary abstract. This abstract describes the problem, the motivation and the proposed approach. It reports no results, since no experiments have been executed at this stage.**

[TO BE COMPLETED: 1–2 sentences on the problem and why it matters] [TO BE COMPLETED: 1 sentence on the gap identified in the literature] [TO BE COMPLETED: 1–2 sentences on the proposed approach and track] [TO BE COMPLETED: 1 sentence on the evaluation protocol and the literature baseline the work aims to match or exceed]

***Index Terms*—[TO BE COMPLETED: keyword], [TO BE COMPLETED: keyword], [TO BE COMPLETED: keyword], [TO BE COMPLETED: keyword], [TO BE COMPLETED: keyword]**

I. Introduction

A. Context and Relevance

[TO BE COMPLETED: 2–3 paragraphs. What is the problem domain, why it matters technically and practically, and what is the concrete cost of not solving it. Cite domain literature.]

B. Problem Statement

[TO BE COMPLETED: 1–2 paragraphs. The specific limitation this work attacks, narrowed from the general domain.]

C. Research Question

> **RQ: [TO BE COMPLETED: One explicit question. Must be answerable with experimental evidence by the end of the semester and must admit refutation.]**

D. Working Hypothesis

> **H: [TO BE COMPLETED: Falsifiable statement with a direction and a magnitude, e.g. "approach X improves metric M by at least D over baseline B under protocol P."]**

E. Objectives

General objective. [TO BE COMPLETED: One sentence.]

Specific objectives.

1) **[TO BE COMPLETED: Objective 1]**
2) **[TO BE COMPLETED: Objective 2]**
3) **[TO BE COMPLETED: Objective 3]**
4) **[TO BE COMPLETED: Objective 4]**

F. Expected Contribution

[TO BE COMPLETED: What the reader gains that does not exist today: a comparison, a formulation, a reproducible pipeline, an agent integration.]

G. Document Structure

Section II reviews the literature along three axes. Section III presents the proposed methodology. Section IV details the technical analysis of the selected track. Section V discusses preliminary ethical considerations. Section VI declares the use of AI tools. Section VII closes with the current state and next steps.

II. Related Work

[TO BE COMPLETED: One short paragraph stating the review methodology: databases queried, search strings, inclusion and exclusion criteria, time window, and how many works were retained out of how many screened.]

A. Axis A: The Problem Domain

How has this problem been addressed in the literature? Which approaches dominate? What is considered solved and what remains open?

[TO BE COMPLETED: 3–5 paragraphs synthesizing at least 7 peer-reviewed works. Group by approach family, not one paragraph per paper.]

[TO BE COMPLETED: Closing paragraph: which approaches dominate, what is settled, what is still open.]

B. Axis B: Techniques of the Selected Track

What is the current state of the methods we intend to use? What are their strengths, limits, and the assumptions under which they work?

[TO BE COMPLETED: 3–5 paragraphs synthesizing at least 7 peer-reviewed works on the technique family of the chosen track.]

[TO BE COMPLETED: Closing paragraph: the technical justification of the track, supported by literature and not by preference. State explicitly which alternative track or technique family was discarded and why.]

C. Axis C: Dataset / Environment

Which works have used it? Which metrics do they report and under which protocol? What is the best published result? Which known problems does it have?

[TO BE COMPLETED: 3–5 paragraphs synthesizing at least 7 peer-reviewed works that use the selected dataset or environment.]

[TO BE COMPLETED: Documented known problems: bias, leakage, size, representativeness. For an RL environment: version differences, reward scaling, and non-comparable evaluation protocols.]

D. Comparative Analysis

[TO BE COMPLETED: 2–3 paragraphs reading the table: which trends are visible, which comparisons are unfair and why, which limitation recurs across works.]

E. Literature Baseline

[TO BE COMPLETED: State the best published result on the selected dataset or environment under the selected metric, with its reference. If a direct comparison is impossible, state a defensible range instead.]

> **Target: [TO BE COMPLETED: The number this work aims to match or exceed, with its metric and its source.]**

Protocol differences that block a fair comparison.

- **[TO BE COMPLETED: Different train/test split]**
- **[TO BE COMPLETED: Different metric or metric variant]**
- **[TO BE COMPLETED: Different data subset or environment version]**
- **[TO BE COMPLETED: Different amount of training budget or compute]**

F. Identified Gaps

Gaps found in the literature.

1) **[TO BE COMPLETED: Gap 1]**
2) **[TO BE COMPLETED: Gap 2]**
3) **[TO BE COMPLETED: Gap 3]**

Gaps this work addresses. [TO BE COMPLETED: Which of the above, and how.]

Gaps out of scope. [TO BE COMPLETED: Which ones remain unaddressed and why they exceed the scope of this project.]

Figure 1. Proposed pipeline. **[TO BE COMPLETED: caption]**

III. Proposed Methodology

A. Problem Formalization

[TO BE COMPLETED: Mathematical formulation. Define the input space, the output space, the objective function, and the constraints. For Track C, point to the MDP in Section IV and state the optimization objective here.]

$$\textbf{[TO BE COMPLETED: objective function]} \tag{1}$$

B. Proposed Pipeline

[TO BE COMPLETED: End-to-end description: data ingestion, preprocessing, model, training loop, evaluation, and the interface the Stage 3 agent will consume.]

C. Experimental Protocol

1) Data Partitioning: **[TO BE COMPLETED: Train / validation / test split with exact proportions. State whether the split is random, stratified, temporal or by group, and justify the choice against the alternatives.]**

2) Validation Strategy: **[TO BE COMPLETED: Holdout, k-fold, nested CV, or episodic evaluation over N seeds. Justify. State how hyperparameters are selected without touching test.]**

3) Metrics:

4) Data Leakage Prevention: **[TO BE COMPLETED: Name every leakage vector considered (target leakage, test set touched during tuning, temporal leakage, group leakage across splits) and the concrete measure taken against each one.]**

D. Baseline Plan

1) **Trivial baseline. [TO BE COMPLETED: e.g. majority class, random policy, rule-based controller.]**
2) **Competitive classical baseline. [TO BE COMPLETED: The strongest non-deep or non-RL method reasonably applicable.]**
3) **Literature baseline.** The published target declared in Section II-E.

E. Reproducibility Contract

[TO BE COMPLETED: Fixed seeds and where they live, saved configuration files, run logging, hardware used, dependency pinning, and the exact command that reproduces each reported number.]

Table I
COMPARATIVE SUMMARY OF RELATED WORK.

Reference	Year	Technique	Dataset / Environment	Main metric	Reported result	Identified limitation
[1]	TBD	TBD	TBD	TBD	TBD	TBD
[2]	TBD	TBD	TBD	TBD	TBD	TBD
[?]	TBD	TBD	TBD	TBD	TBD	TBD
[?]	TBD	TBD	TBD	TBD	TBD	TBD
[?]	TBD	TBD	TBD	TBD	TBD	TBD

Table II
EVALUATION METRICS AND THEIR JUSTIFICATION.

Metric	Why this metric, and what it beats
[TO BE COMPLETED: primary]	**[TO BE COMPLETED: justification]**
[TO BE COMPLETED: secondary]	**[TO BE COMPLETED: justification]**
[TO BE COMPLETED: secondary]	**[TO BE COMPLETED: justification]**

Table III
IDENTIFIED RISKS AND THEIR MITIGATION.

Risk	Impact	Mitigation / trigger
[TO BE COMPLETED: risk]	**[TO BE COMPLETED: H/M/L]**	**[TO BE COMPLETED: mitigation]**
[TO BE COMPLETED: risk]	**[TO BE COMPLETED: H/M/L]**	**[TO BE COMPLETED: mitigation]**
[TO BE COMPLETED: risk]	**[TO BE COMPLETED: H/M/L]**	**[TO BE COMPLETED: mitigation]**

F. Risks and Mitigation

G. Planned Agent Integration

[TO BE COMPLETED: How the trained model will be consumed by an intelligent agent in Stage 3: LLM with tool calling, RAG system, multi-agent architecture, or an RL agent operating in its environment. State the interface and how the consumption will be verified.]

IV. TECHNICAL ANALYSIS OF THE SELECTED TRACK

[TO BE COMPLETED: One sentence naming the approved track and the date of approval.]

A. Exploratory Data Analysis

[TO BE COMPLETED: Number of records, number of features, target distribution, temporal coverage.]

1) Data Dictionary:

2) Data Quality and Imputation: **[TO BE COMPLETED: Missing value pattern and the justified imputation strategy.]**

3) Outlier Detection: **[TO BE COMPLETED: Method, threshold, and what is done with the detected outliers.]**

Table IV
DATA DICTIONARY.

Variable	Type	Meaning / range / missing %
[TO BE COMPLETED: var]	**[TO BE COMPLETED: type]**	**[TO BE COMPLETED: description]**

4) Class Imbalance: **[TO BE COMPLETED: Ratio, and the strategy chosen with its justification.]**

B. Corpus / Image Set Analysis

[TO BE COMPLETED: Size, class distribution, sequence length or resolution distribution, documented biases of the source.]

1) Preprocessing and Tokenization: **[TO BE COMPLETED: Pipeline and justification.]**

2) Data Augmentation: **[TO BE COMPLETED: Transformations, why each one is valid for this domain, and which were discarded as label-destroying.]**

3) Hardware and Memory Considerations: **[TO BE COMPLETED: Model size, batch size, precision, expected GPU memory, and estimated training time on Google Colab Pro.]**

C. MDP Formalization

The task is modeled as a Markov decision process $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$.

1) State Space $\mathcal{S}$: **[TO BE COMPLETED: Exact variables, ranges, normalization, and whether the environment is fully or partially observable.]**

2) Action Space $\mathcal{A}$: **[TO BE COMPLETED: Discrete or continuous, dimensionality, bounds.]**

3) Transition Dynamics $P(s' \mid s, a)$: **[TO BE COMPLETED: Simulator, its version, stochasticity sources, timestep duration, and episode termination conditions.]**

4) Reward Function $R(s, a, s')$:

$$R(s, a, s') = \textbf{[TO BE COMPLETED: reward expression]} \quad (2)$$

[TO BE COMPLETED: Justify every term and every weight. State which reward designs were discarded and which degenerate behavior each one would induce.]

5) Discount Factor γ: **[TO BE COMPLETED: Value and justification with respect to the effective horizon.]**

Table V
USE OF AI TOOLS PER TEAM MEMBER.

Member	Tool	Where, purpose, and what was manually verified
A. Chaves	**[TO BE COMPLETED: tool]**	**[TO BE COMPLETED: section, purpose, verification]**
D. Duarte	**[TO BE COMPLETED: tool]**	**[TO BE COMPLETED: section, purpose, verification]**
S. Hernández	**[TO BE COMPLETED: tool]**	**[TO BE COMPLETED: section, purpose, verification]**

6) Environment Verification: **[TO BE COMPLETED: Smoke test executed, random-policy return, and confirmation that the environment runs on Google Colab Pro.]**

V. Preliminary Ethical Considerations

A. Data Sensitivity

[TO BE COMPLETED: Does the dataset contain personal, sensitive or protected data? Under which license was it released and what does that license permit?]

B. Bias Risk

[TO BE COMPLETED: Which populations, conditions or regimes are under-represented, and what failure this could cause downstream.]

C. Mitigation Planned Before Training

- **[TO BE COMPLETED: Mitigation 1]**
- **[TO BE COMPLETED: Mitigation 2]**

D. Deferred to Stage 3

Deployment ethics, misuse scenarios and the operational risks of the intelligent agent are addressed in the final stage.

VI. Declaration of AI Tool Usage

Manual verification performed. **[TO BE COMPLETED: e.g. every DOI resolved manually in IEEE Xplore or ACM DL; every reported metric checked against the source PDF; every generated snippet executed and inspected.]**
Not AI-generated. The research question, the problem formalization, **[TO BE COMPLETED: the MDP / the experimental protocol]**, and all experimental results are the authors' own work.

VII. Current State and Next Steps

[TO BE COMPLETED: 2–3 paragraphs: what the review established, which gap the work attacks, and which number it aims to beat.]
Next steps (Stage 2).

- **[TO BE COMPLETED: Implement the pipeline and the trivial baseline]**
- **[TO BE COMPLETED: Implement the competitive classical baseline]**
- **[TO BE COMPLETED: Train the main model and run the ablations]**
- **[TO BE COMPLETED: Compare against the three references under the declared protocol]**

References

[1] T. B. COMPLETED, “To be completed,” *TO BE COMPLETED*.
[2] ——, “To be completed,” in *TO BE COMPLETED*.